Question-design rules for the next mocks

Version 2. Part I keeps the option rules from the first draft. Part II adds what the research says actually drives difficulty in passage-based reasoning items, with the levers I will pull while writing.

Part I. The option rules (carried forward)

Item-writing research is clear that extreme words are a defect rather than a difficulty: specific determiners such as never and always are generally wrong answers and can be eliminated without reading the passage, and vague hedges such as usually and may are generally right answers. The first version of these rules therefore banned both, and pushed difficulty into the relationship between options. That holds, with one correction set out in section 2.5 below.

R1. No specific determiners, anywhere. No all, always, never, only, solely, must, entirely, none, cannot, at all costs - in the key or in a distractor.

R2. No hedge tells either. No usually, typically, may, can, sometimes, tends to, in most cases, used as a marker of correctness. All four options carry comparable modal strength, so no option can be chosen or rejected on strength alone.

R3. Equal architecture. Options sit within about twenty per cent of each other in length, share the same grammatical shape, and carry the same number of clauses.

R4. The key is never the odd one out. Not the longest, not the shortest, not the most qualified, not the only option repeating a term from the passage.

R5. Four named species of distractor. The exact negation of the key; a scope shift; a premise booster; or a half-right option whose second clause fails.

R6. No single-word eliminations. Read each option alone and ask what word would let a candidate discard it. If there is one, rewrite.

R7. Two options must survive a first reading. In at least one question in five, the two strongest options both remain defensible to a careful but not close reader.

Part II. What actually makes a reasoning item hard

2.1 The evidence, in order of how much it matters

Four findings reorder my instincts about difficulty. First, the passage is not where most of the difficulty lives. A study of crowdsourced questions across seven passage sources found that passage source, passage length and readability measures did not significantly affect question difficulty; what did was the type of reasoning the question demanded, with hard items requiring multi-sentence reasoning and knowledge inference. A separate analysis of item difficulty found the decision process - mapping each option back to the text and judging it - mattered more than the difficulty of the text itself.

Second, the options are where difficulty lives, and two features dominate. Distractors written in the same structure as the key raised mean difficulty substantially in a controlled study - about twelve points in pass rate - with everything else held constant. And each additional plausible distractor adds roughly thirteen per cent to the proportion of wrong answers. Four working distractors, not two, is the target.

Third, within the passage, specific features carry measurable cost. Studies of reading items find that negations increase processing time; that a high density of referential expressions - pronouns and definite references that must be resolved - makes comprehension harder; that fronted structures, meaning sentences that open with a subordinate clause or a cleft, make main-idea questions harder; and that information placed in the middle of a passage or paragraph is harder to locate than information at the start or end. Propositional density, syntax and word difficulty predict item difficulty; raw word count does not.

Fourth, the cognitive literature supplies one deliberate hazard. Interesting but irrelevant details - the seductive details effect - damage comprehension in three ways: they distract attention, they break the coherence of the argument so the reader has to repair it, and they divert the reader into building a model around the wrong material. That is a legitimate difficulty lever, used sparingly, because it also seeds a very good distractor.

2.2 Levers in the passage

Passage length stays where it is. The work is done by density and structure, not by adding words.

Lever	Mechanism	How I will use it
Propositional density	More claims per sentence, with the relation between them implied rather than spelled out	Two or three propositions per sentence; cut every sentence that carries only one
Fronted structures	Subordinate clauses and clefts at the head of a sentence raise the cost of finding the main idea	Two or three per passage, always on a sentence the questions depend on
Referential chains	Pronouns and definite references must be resolved to follow the argument	Six to ten per passage, with at least one chain running across a paragraph break
Negation	Negated propositions cost processing time	Two or three per passage, placed where they change the sense of a claim
Buried load-bearing sentence	Information in the middle of a paragraph is harder to find than at its edges	The sentence a question turns on sits mid-paragraph, never first or last
Coherence break	Omitted connectives force the reader to supply the causal or adversative link	Two places per passage where the relation between adjacent claims is left unstated
Seductive detail	Interesting, vivid, irrelevant material diverts and disrupts	One per passage, at most, and never load-bearing
Withheld stance	An author's view expressed only through evaluative adjectives forces a model of the author	At least one passage per paper with no thesis sentence
Attributed claims	Views belonging to critics or officials sit next to the author's own	Every passage carries at least one claim that is not the author's

2.3 Levers in the question

Lever	Mechanism	How I will use it
Evidence spread	Requiring two or three locations raises the mapping cost more than any wording change	Every core question joins at least two locations; one per set joins three
Inference distance	The correct answer is the smallest defensible step beyond the text, not a restatement	No answer is a paraphrase of a single sentence
Synthesis across paragraphs	The hardest inference items combine two parts of the passage into a claim neither states	At least one per set
Function questions	Asking what a sentence or paragraph is doing forces structural reading	One per set: role of a sentence, or purpose of a paragraph
Application and analogy	Abstracting a relation and instantiating it cannot be done by keyword matching	One per two sets
Comparative stems	Most accurately, most strongly supported, the strongest objection - the candidate must rank	The default stem form for inference items
Tension inside the passage	A paradox or point at issue that the passage itself sets up and half resolves	One set per paper built this way
No paragraph pointers	Naming a paragraph lowers difficulty sharply	Only the anchor question in each set names where to look

2.4 Levers in the options

Lever	Mechanism	How I will use it
Structural similarity	Distractors shaped like the key raise difficulty by around twelve points of pass rate	All four options share syntax, clause count and length band
Four working distractors	Each plausible distractor adds about thirteen per cent to wrong answers	No filler option; every wrong option is chosen by someone who half-understands
Species discipline	Wrong options that reproduce real misreadings teach while they test	Negation, scope shift, premise booster, half-right, wrong viewpoint, outside knowledge, too narrow, overreach
Attractive distractor placement	Distance between the strongest distractor and the key affects response behaviour	The strongest distractor sits next to the key; key position still balanced across the paper
Length at the long end	Longer options raise inference difficulty, so the fix is not a short key	All four options long and complex, key near the mean
No convergence	If one option is right for the reason another is right, both collapse	Check that no two options share a supporting premise

2.5 The hedging paradox, and the correction to R2

There is a conflict in the advice, and it needs stating because it is the difference between a hard paper and a paper with a trick. The hardest inference questions in competitive reasoning tests typically have a correct answer whose hedge - tends to, in some cases, generally - is precisely what distinguishes it from a close-looking wrong answer that overstates. So hedging sometimes has to be the discriminator. But if hedged options are correct more often than not, a candidate learns to pick the hedge and stops reasoning.

The resolution is to make hedging uncorrelated with correctness across the paper. Concretely: some keys are hedged and some are flat; some distractors are hedged and some are flat; and where the hedge is the discriminator, a second distractor is also hedged but wrong for a substantive reason, so that the candidate cannot choose on tone. I will count hedged keys and hedged distractors for the whole paper and keep the ratio near one to one. This supersedes the simple ban in R2.

2.6 The limit: reading load is not reasoning load

Every lever above can be overdone, and two of them change what the paper measures. Heavy referential density and long fronted sentences raise the cost of decoding the prose; past a point the item tests patience with syntax rather than reasoning, and a candidate who is a slow reader is punished twice. The same applies to seductive details: they depress comprehension, but they depress it indiscriminately, including for readers who reason well. So the target is abstract content in plain words, with the difficulty carried by the relations between claims and between options - not by obscurity of vocabulary, not by paragraph length, and not by burying the answer in a thicket of pronouns.

2.7 Difficulty budget for each set

A set of six questions will be built to this distribution, so that the paper has a shape rather than a uniform texture. The anchor question is deliberately answerable, and it is the only one allowed to name a paragraph.

Slot	Share of a set	What it tests	Where the answer sits
Anchor	One question	Retrieval or meaning in context	One sentence, locatable
Core	Three questions	Inference, function, strengthen or weaken	Two locations, mid-paragraph
Hard	One question	Synthesis, application or analogy	Three locations, or an unstated authorial view
Wildcard	One question	Flaw, point at issue or resolve the paradox	The tension the passage sets up

2.8 The audit, revised

Check	What it catches
Strength test	Can the answer be predicted from modal strength alone, anywhere in the paper?
Single-word test	Is any option discardable on one word?
Length test	Is the key longer or shorter than the mean of its distractors, in the question or the paper?
Overlap test	Can two options both be right on a fair reading? Are numeric ranges nested?
Species test	Does every distractor belong to one of the named types?
Two-survivor test	Does at least one question in five leave two options standing on a first reading?
Spread test	Does every core question join at least two locations in the passage?
Hedge-ratio test	Are hedged keys and hedged distractors roughly equal in number across the paper?
Convergence test	Do any two options rest on the same supporting premise, so that both collapse together?
Position test	Is the key position balanced, and does the strongest distractor sit next to the key?
Decode test	Would a good reasoner who reads slowly run out of time on this passage rather than get it wrong?

Two of these are new and both are about validity rather than difficulty: the hedge-ratio test, which stops tone becoming the answer, and the decode test, which stops the paper becoming a reading-speed contest. A question that fails any of the first six is rewritten; a passage that fails the decode test is rewritten.